When Can a Higher Loan Spread Predict Lower Delinquency?

Endogenous Screening under Borrower Opacity

Fabián Matías Galarza Chumbe
https://github.com/fgalarzac/ai-project

Topic presentation, October 14, 2026

1 The question and the track

Question. When does a bank's loan rate, measured relative to peers in the same product, maturity, currency, and month, predict lower rather than higher future delinquency? This is a **Track B** project: it supplies the missing agent problem behind my thesis on Peruvian commercial banks. The thesis currently writes the predictive slope as a reduced-form difference between an adverse-selection channel and a screening channel. The adaptation below makes the bank choose screening, derives both channels from primitives, and states the condition under which their ranking reverses with borrower opacity. The object is a forecasting mechanism, not a causal effect of changing a posted rate.

2 The baseline and the model the thesis currently lacks

The ingredients exist separately. In Varian (1980), informed and uninformed buyers generate price dispersion. In Broecker (1990), banks with imperfect credit tests compete in interest rates, and a rate cut can improve the attracted pool. Ruckes (2004) endogenizes screening intensity and obtains an inverse-U response to the prior probability of a good borrower. Dell'Ariccia and Marquez (2006) shows how information asymmetries across banks can switch an equilibrium between screening and pooling. My adaptation combines a relative-price sorting margin with a continuous screening choice and makes borrower opacity the moderator.

Let $D \in \mathcal{D} = [\underline{D}, \overline{D}]$ be the bank's spread relative to its peers and $\kappa \in [0, 1]$ the opacity of the borrower segment. The game has two stages. First, the bank posts D and receives continuous loan demand $L(D, \kappa) > 0$, with $L_D < 0$. Second, after observing the applicant pool, it chooses screening effort $e \in [0, \bar{e}]$. Before screening, the probability of default among potential originations is

$$q^0(D, \kappa) = \bar{q} + a(\kappa)D, \qquad a(\kappa) > 0. \tag{1}$$

The coefficient a is the sorting effect of a relative price: a higher spread loses comparison shoppers and worsens the remaining pool. Screening lowers default according to

$$q(D, e, \kappa) = q^0(D, \kappa) - g(\kappa)e, \tag{2}$$

and costs $c(\kappa)e^2/2$ per loan. A performing loan pays $R(D) = \bar{R} + D > 0$, while F collects funding and operating costs that do not depend on screening. The bank's problem is

$$\max_{D \in \mathcal{D}} \; L(D, \kappa) \left\{ \max_{e \in [0, \bar{e}]} [1 - q^0(D, \kappa) + g(\kappa)e]R(D) - F - \frac{c(\kappa)}{2}e^2 \right\}. \tag{3}$$

Compactness and continuity give a first-stage solution. The proposed result is about the continuation quality map $q^*(D, \kappa)$, which is defined for any observed relative price, including the equilibrium price and price variation generated by bank-specific cost or markup shocks. This keeps the forecasting claim separate from a causal comparative static of D^*.

3 The result I expect

For an interior screening choice, the first- and second-order conditions are

$$g(\kappa)R(D) - c(\kappa)e = 0, \qquad \frac{\partial^2 \pi}{\partial e^2} = -c(\kappa) < 0. \tag{4}$$

Thus

$$e^*(D, \kappa) = \frac{g(\kappa)}{c(\kappa)}(\bar{R} + D), \quad q^*(D, \kappa) = \bar{q} + a(\kappa)D - h(\kappa)(\bar{R} + D), \quad h(\kappa) \equiv \frac{g(\kappa)^2}{c(\kappa)}. \tag{5}$$

The new object h is *screening leverage*: the effect on portfolio quality of the screening response induced by a larger loan payoff.

Candidate Proposition 1 (opacity threshold). Suppose: (i) the solution is interior and $q^* \in (0, 1)$; (ii) a and h are continuous; (iii) $a'(\kappa) < 0$; (iv) $h'(\kappa) > 0$, equivalently $2g'(\kappa)/g(\kappa) > c'(\kappa)/c(\kappa)$; and (v) $a(0) > h(0)$ while $a(1) < h(1)$. Then there is a unique $\kappa^* \in (0, 1)$ satisfying $a(\kappa^*) = h(\kappa^*)$, and

$$\beta(\kappa) \equiv \frac{\partial q^*}{\partial D} = a(\kappa) - h(\kappa) \begin{cases} > 0, & \kappa < \kappa^*, \\ = 0, & \kappa = \kappa^*, \\ < 0, & \kappa > \kappa^*. \end{cases} \qquad \frac{\partial^2 q^*}{\partial D \partial \kappa} = a' - h' < 0. \tag{6}$$

The proof is the intermediate value theorem plus strict monotonicity. If future delinquency satisfies $m_{t+s} = \rho_s q_t^* + \varepsilon_{t+s}$, with $\rho_s > 0$, its predictive slope has the same sign. Importantly, higher screening cost alone makes h smaller and cannot deliver the reversal. The economic restriction is that opacity raises the marginal information gain enough to dominate any increase in cost. This is a falsifiable sufficient condition, not a definition disguised as a result.

4 Why this is not already solved

I checked the model and appendices of the accessible 2004 version of Ruckes (2004), the 2005 working-paper record and the full IMF reprint of Dell'Ariccia and Marquez (2006), and the stated equilibrium result of Broecker (1990). Broecker fixes the testing technology and studies how rate competition sorts borrowers. Ruckes solves for screening intensity and mixed loan pricing as economic prospects change, but does not characterize the sign of relative price as a default signal across opacity segments. Dell'Ariccia and Marquez vary information asymmetry across banks and derive screening-versus-pooling regimes, not a continuous price-to-quality derivative. Varian supplies the shopper/captive-buyer logic but has neither screening nor default. The proposed derivative $a(\kappa) - h(\kappa)$, its threshold, and the distinction between costly screening and valuable screening therefore do not appear in those baselines.

5 Plan and risks

I will prove Proposition 1 and a robustness version with a general increasing repayment function $R(D)$, simulate the full parameter region rather than a single calibration, and formalize the threshold theorem in Lean after freezing the paper proposition. The code already checks the FOC, SOC, crossing, monotonicity, interiority, and probability bounds for a transparent calibration. The empirical thesis will map the theoretical κ to pre-specified segment-level opacity proxies and test out-of-sample forecasts; it will not label the estimated coefficient causal.

The main risk is condition (iv). If the data or a richer signal technology do not support increasing screening leverage, I will not force a sign reversal. The fallback result will characterize three regions, $a > h$, $a = h$, and $a < h$, and treat the existence of a threshold as an estimable restriction. A second risk is an endogenous-price confound; the fallback is to keep the theorem explicitly conditional on observed D, add bank and segment-by-month controls, and interpret the exercise only as forecasting.

References

Thorsten Broecker. Credit-worthiness tests and interbank competition. *Econometrica*, 58(2):429–452, 1990. doi: 10.2307/2938210.

Giovanni Dell'Ariccia and Robert Marquez. Lending booms and lending standards. *The Journal of Finance*, 61(5):2511–2546, 2006. doi: 10.1111/j.1540-6261.2006.01065.x.

Martin Ruckes. Bank competition and credit standards. *The Review of Financial Studies*, 17(4):1073–1102, 2004. doi: 10.1093/rfs/hhh011.

Hal R. Varian. A model of sales. *American Economic Review*, 70(4):651–659, 1980.